

Guide: Certified Merge Progressions

Collatz proof project — September 5, 2026

What is now verified

Lean checks 360 arithmetic rules. Each gives a smaller positive seed that meets the original orbit and inherits full noncontraction if the original has it. Every rule applies to all nonnegative quotients, so it covers an infinite progression, not just a tested example.

What noncontraction means

After t shortcut steps with j odd steps, the multiplicative coefficient is $3^j/2^t$. A seed is noncontracting when that coefficient is at least one for every prefix. No infinite noncontracting seed has been constructed. The aim is to rule out a least positive noncontracting seed by finding a smaller one whenever the original has that property.

Why the transfer works

Before the meeting, the replacement's coefficient must stay at least one. At the meeting, it must be at least as large as the original path's coefficient. Afterward the paths follow the same values and gain the same further factors. Those two checks preserve full noncontraction. Shifting the original seed does not automatically preserve noncontraction; the coefficient comparison is essential.

An example that works for arbitrarily large seeds

For any whole number $Q \geq 0$, take

$$N = 111 + 4374Q, \quad M = 103 + 4096Q.$$

The larger seed reaches $167 + 6561Q$ in one step; the smaller reaches it in twelve steps with eight odd steps. The difference $N - M = 8 + 278Q$ stays positive. Thus a least positive noncontracting seed cannot belong to this progression. This does not, by itself, prove that every member converges.

How a finite certificate gives an infinite rule

Adding a multiple of 2^t to a seed preserves its first t parities. The endpoint changes by a corresponding multiple of 3^j . The rule chooses increments for the two seeds so these endpoint changes agree, while the replacement's increment is no larger. Parity validity, merging, and strict seed ordering then persist for every nonnegative quotient.

The kernel checks the actual endpoints, prefix inequalities, coefficient comparison, and balanced increments for every listed rule. A general Lean theorem handles every quotient. Python generates the data; it does not supply a trusted proof.

What remains open

The 3,768 successful dominating certificates from the depth-18 search reduce to 360 distinct progressions. Their validity is kernel certified; the search's completeness is not. The earlier bounded search still leaves 2,336 seeds unresolved, including 27 and 703; they are not counterexamples.

Most importantly, no theorem says that a suitable rule exists for every hypothetical least noncontracting seed. Proving that coverage, potentially with lengths depending on the seed, remains the central gap. Nontrivial cycles are also not excluded here.

Reproduce

Run `python3 analysis/merge_progressions.py --write-lean` from the root and `lake build Collatz.MergeRuleTable` from `lean/`. The saved JSON, generated Lean table, and tests make the rules reviewable. The Collatz conjecture remains unproved.